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Exploring biosorption properties of *Litchi chinensis* peel for a cationic dye Rhodamine 6G in Liquid/Solid phase system: Kinetics and equilibrium studies

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ABSTRACT

The Rhodamine 6G (Rh6G), a cationic dye is actively being used as a potentially tracing dye in the field of biotechnology for immunological studies. The peel powder of *Litchi Chinensis* was employed in native condition to explore the sorption properties at different experimental conditions like initial dye concentration, contact time, dosage of biosorbent, temperature and pH. The detailed characteristics of *Litchi chinensis* peel before and after adsorption were performed by using sophisticated instruments like SEM-EDX, FTIR, BET and XRD. From the adsorption studies it was observed that the equilibrium data followed Pseudo-second order rate kinetics and Freundlich isotherm. The Langmuir adsorption capacity (Q_o) was found to be 6.66 mg/g, and the adsorption process was spontaneous and exothermic in nature (based on ΔG, ΔH and ΔS values). All the laboratory experiments proven to be an evident for the *Litchi chinensis* peel as an efficient biosorbent for the removal of Rhodamine 6G in the liquid/solid phase system.

Introduction

Water pollution is the major problem on the earth. Its effect mainly plants, animals, human beings and environment too. Industrial and pharmaceutical waste produces toxic pollutants. Such as heavy metals, dyes, radio nucleotides, rare earth elements, etc. Commonly dyes have complex properties such as high stability and aromatic structure. Dyes are an important class of pollutants that are present in large amounts in waste waters from textile, dyeing, cosmetic, petroleum, leather, paper and paint industries. These industries produce millions of contaminated wastewater every day, which, when not properly treated and discharged dye effluents directly into the fresh river water, can cause serious environmental contamination (Silva et al., 2020). Dye removal processes were biological oxidation, chemical precipitation, photo catalytic degradation, coagulation, flocculation and adsorption. Among all these processes, adsorption is considered to be the most essential and suitable process. On comparison with other techniques adsorption technique has low cost, high removal capacity, simple, ability to separate several chemical compounds and lower sensitivity to the toxic agents of the waste water. Adsorption explained the process of transferring liquid to solid, resulting in either physical or chemical interaction (Silva et al., 2020). Rhodamine 6G (Rd6G) was soluble in water which is a synthetic dye mainly used in textiles of cotton, wool, silk and food stuffs and was used as a colorant (Li et al., 2018). Shen and Gondal (2017) used in ELISA test and forensic department (Thitipone et al., 2020). The Rd6G is highly toxic in nature and causes harmful effects such as allergic dermatitis, eye irritation, respiratory problems, neurotoxicity reproductive etc., (Shen and Gondal, 2017) and developmental issues and cancer to human beings (Umapada et al., 2016). Apart from these, Rd6G contaminated drinking water can also leads to subcutaneous tissue borne sarcoma, which is highly carcinogenic (Umapada et al., 2016). Now a day's removal of dyes, heavy metals and ions are using agricultural wastes or byproducts – such as pomegranate peel (Nehaba 2017), garlic peel (Gilani et al., 2020), banana peel (Mondal and Kar, 2018) orange peel (Daouda et al., 2019), lemon peel (Ahmed and Alam 2016) etc. In the present study, *Litchi chinensis* peel was used to remove the Rd6G. *Litchi chinensis* belongs to *Sapindaceae* family and an ever green tree having a very popular fruit and widely consumed. Its cultivation is mainly in the east and also spread to African and American countries. It is a nutrient rich fruit. Many researchers reported that *Litchi chinensis* peel and seed were used as an adsorbent for the removal of dyes and heavy metals (Aniline blue, Acid blue 25 dye, Congored, Cr (VI) and Cadmium (Cd)) from waste water.

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Materials and methods

Adsorbate preparation

Rhodamine 6 G (Rd6G) is a cationic dye has a dark reddish purple color. Its molecular formula is $C_{28}H_{31}ClN_2O_3$ and molecular weight is 479.02 g mol^{-1}. It is purchased from Sd fine chem Limited. Preparation of stock solution, 1.0 g of Rhodamine 6G was dissolved in one liter of distilled water. The prepared solution of Rd6G has pH 4.7 so it was adjusted to pH 8 using 0.1 N HCL and 0.1 N NaOH. The constant pH 5 was maintained throughout all experiments.

Adsorbent preparation

Litchi chinensis fruit was purchased from local market in Kadapa, Andhra Pradesh, India. The litchi fruits peel which was a waste is used for the preparation of adsorbent. The obtained peel was washed with distilled water thrice to remove dust particles and dried it in hot air oven at 75 °C for 48 hrs. The peel was grinded and sieved with 100 μm sieve. Finally, the dried powered was used as an adsorbent.

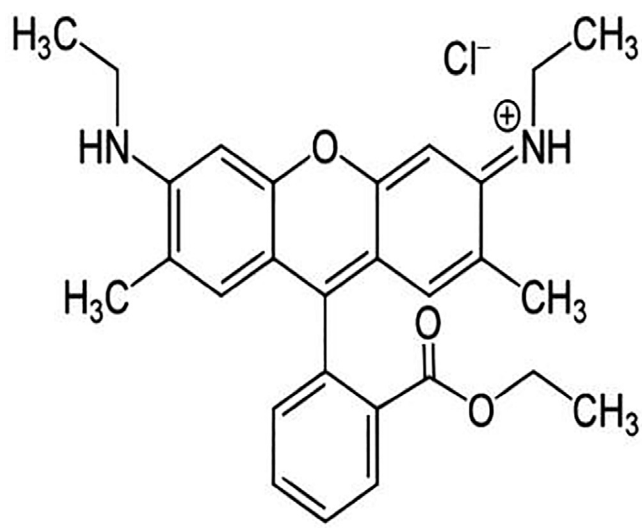


Fig. 1. The structure of rhodamine 6 G.

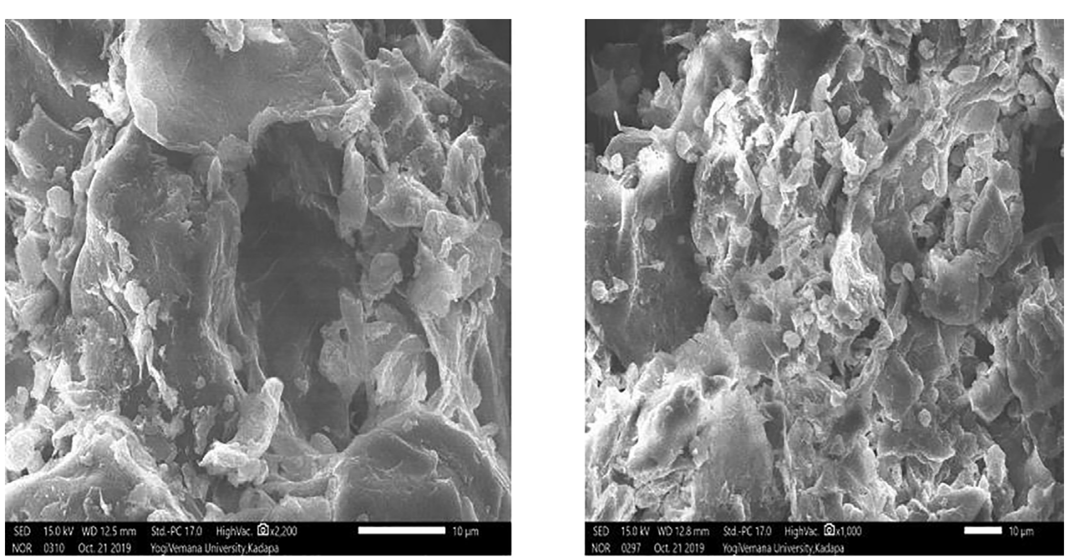
Batch adsorption

The batch adsorption study was conducted in 250 ml conical flasks filled with 50 ml of adsorbate at different initial concentrations of 20–50 mg/L for R6G. Dose of 300 mg was added into the solution and agitated at 180 rpm using rotary shaker. The solution was separated from the adsorbent by the centrifugation at 4500 rpm for 20 min. The absorbance of dye was measured by a double beam UV– Vis Spectrophotometer (Lab India UV 3000 +) at 528 nm for Rd6G (See Fig. 1).

Result and discussion

Instrumentation

SEM micrographs were compared with Rd6G dye loaded *Litchi chinensis* adsorbent. The micrographs showed *Litchi chinensis* peel powder surface have rough and porous before adsorption, so here is a good possibility for the dye to be absorbed into these pores. After adsorption the surface becomes smooth and surface of the adsorbent has covered with dye molecules, these differences in the pores and surface irregularities of adsorbent due to the presence of dye molecules (Fig. 2a, b). The EDX spectra revealed the presence of carbon (C), oxygen (O_2), Magnesium (Mg) and Potassium (K) on both loaded as well as unloaded surface of the adsorbent samples which may play a possible role in the process of adsorption of dye molecules (Fig. 3a, b). The availability and active participation of the functional groups on the surface of the *Litchi chinensis* peel were determined by FTIR spectral studies (Fig. 4a, b). The band at 3654 and 1639 cm^{-1} are correspond to the O—H bond vibrations. The peaks at 2934 cm^{-1} represents the stretching of C—H bonds in the CH_2 and CH_3 groups, and the peaks at 2132 cm^{-1} designates the presence of the CO_2 in the surroundings of the sample which can be nullify by baseline correction. The band at 1737 cm^{-1} was assigned for the C=O bending vibrations



(a) (b)

Fig. 2. SEM images of *Litchi chinensis* (a) Before adsorption (b) After adsorption of.

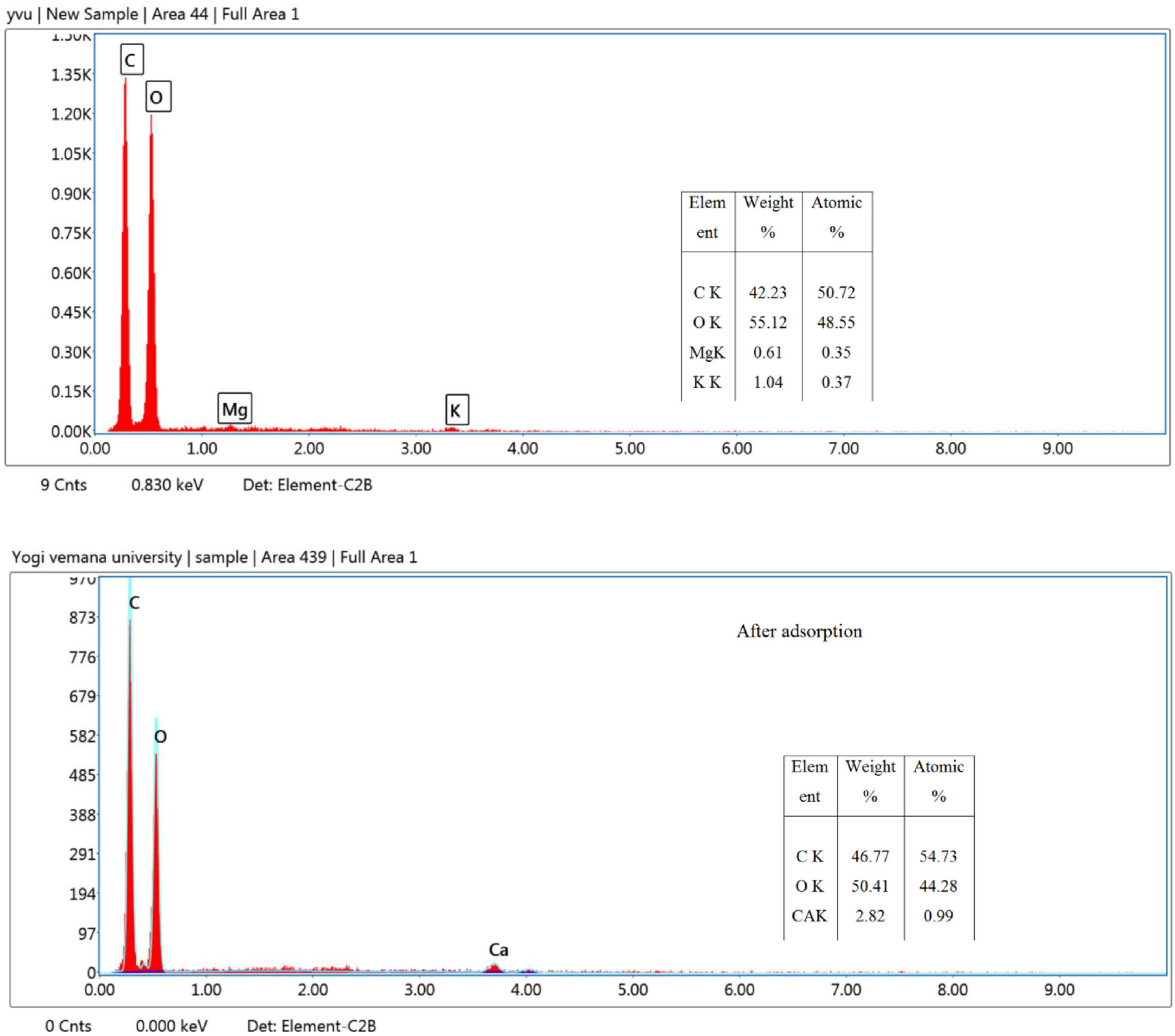


Fig. 3. EDX images. (a) Before adsorption (b) After adsorption of Rhodamine 6G.

in the –COOH (carboxylic acid) groups and the peaks at 1460 and 1282 cm^{-1} indicates the C—H—C bond vibration and the presence of methylene groups respectively. The main modifications are identified as the peaks shifted from 3654 to 3647 cm^{-1} and 1639 to 1647 cm^{-1} corresponds to the involvement of O—H functional groups, the peaks shifted from 2924 to 2942 cm^{-1} and 1737 to 1748 cm^{-1} were corresponds to the participation of C—H and C═O bonds. The C—H bond shift was observed from 2850 to 2844 cm^{-1} due to the transmittance changes which indicate the occurrence of methoxyl group as a result of the removal of lignin and lipids (Adeleke et al. 2019). These observations were depicted that the O—H, carboxyl and carbonyl groups played influencing role in the adsorption and showed the profound effects on the surface modification on the *Litchi chinensis* peel with Rd6G. The Fig. 5a, b represent two diffractogram where loaded *Litchi chinensis* adsorbent was compared with raw adsorbent. The attainment of differential peaks with spinal structures corresponds to the successful grafting of dye molecules on the surface of the adsorbent before adsorption. The obtained sharp peak 2θ at the intensity of 22.079 and 24.433, 22.212 and 24.433 are referred to the presence of RhD6G dye molecules (Fig. 5a, b). Hence, it can be obtained that this was due to the adsorption occurred onto the surface of the *Litchi chinensis* peel powder.

Effect of dye initial concentration and contact time

The adsorption of Rd6G by *Litchi chinensis* peel powder in time with the initial concentration was studied. The experimental results described that adsorption of dye onto the adsorbent material increased with time and the equilibrium was found to be 20, 30, 40 and 60 min for 20, 30, 40 and 50 mg/L of dye concentration respectively, while the uptake was increased at equilibrium from 3.0 mg/g to 5.8 mg/g with an increase of concentration from 20 mg/L to 50 mg/L (Fig. 6). These findings are in accordance with the results observed by Magnetic grapheme oxide (Gautam et al., 2020), when the efficiency of the Rd6G adsorption on to the Magnetic Graphene oxide rapidly increased in the initial stage of contact time and remained unchanged after 30 min of adsorption (Bhullar et al., 2018), has observed that 60 % of Rd6G dye removal was achieved in first hour of the experiment by composite hydrogel, then after the removal became slow and took 8 hrs to reach 90 % of dye removal. This might be because of the freely available active sites at in the initial stages of adsorption, while, as time increases after 60 min the adsorption became resistance due to low binding sites. Similar findings are noticed in *Trichoderma harzianum* mycelia waste (Sadhasivam et al., 2007), *Centella asiatica* (Nirmala et al., 2016).

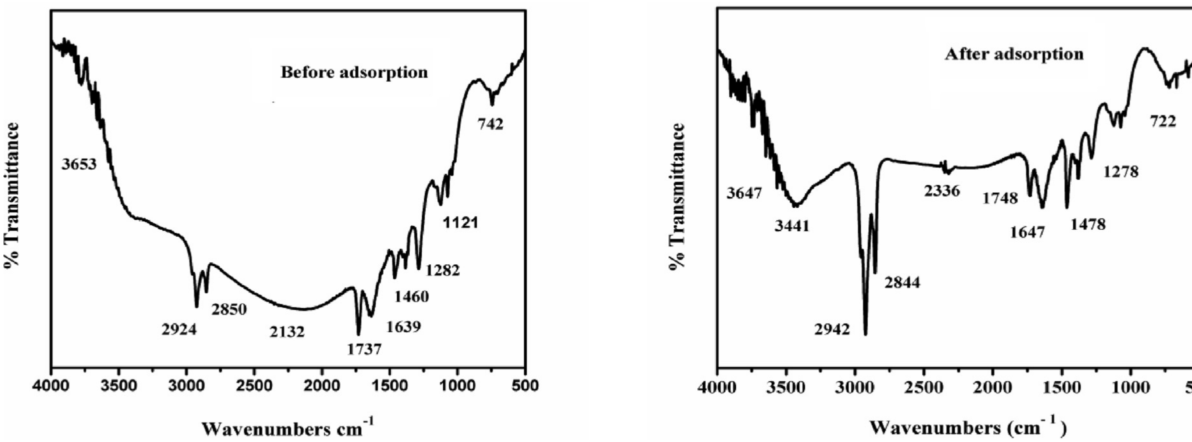


Fig. 4. FTIR images. (a)Before adsorption (b) After adsorption of Rhodamine 6G.

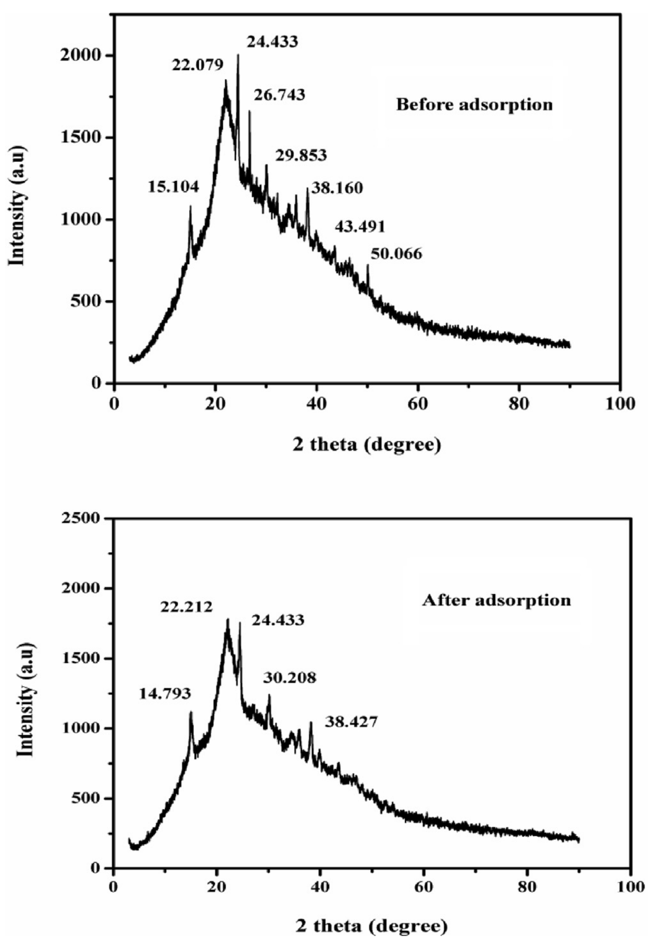


Fig. 5. XRD images. (a)Before adsorption (b) After adsorption of Rhodamine 6G.

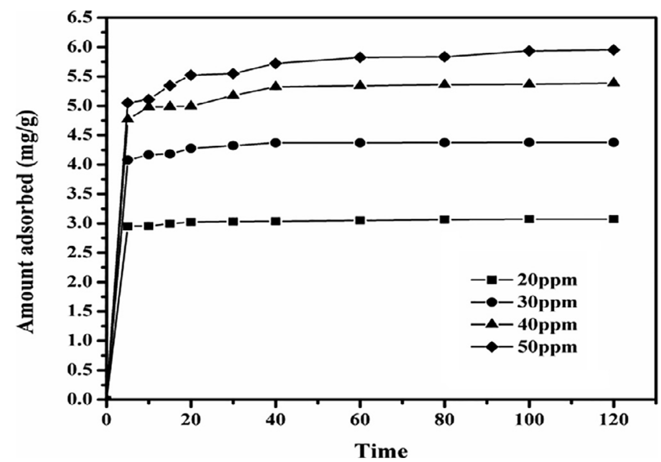


Fig. 6. Effect of contact time on Rhodamine 6G.

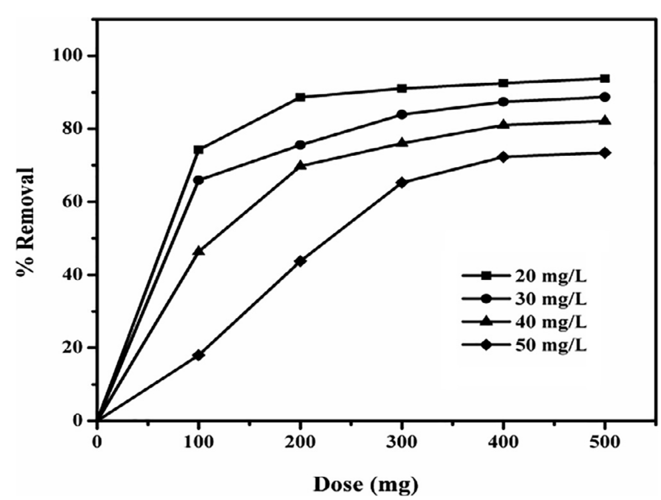


Fig. 7. Effect of dose on Rhodamine 6G.

Effect of adsorbent dose on removal of Rhodamine 6 g

Adsorption investigations were arrangement with various amounts of *Litchi chinensis* peel powder (100, 200, 300, 400 and 500 mg/50 ml) and Rd6G concentration 20 to 50 mg/L. The percent removal of Rd6G increased with the increase in adsorbent dose and reached a persistent value after a particular adsorbent dose (Fig. 7). A larger mass of *Litchi chinensis* could adsorb larger amount of Rd6G because of the availability of extra surface area of the adsorbent, maximum removal of Rd6G were found to be around 93.83 % for the *Litchi chinensis* for the initial concentration of 20 mg/L. Similar results are also observed in biopolymer-based composite hydrogel (Navneet and Kaur, 2017), *Litchi chinensis* (Sirisha et al., 2019).

Effect of temperature

In this study, it was examined that temperature effects on interaction of adsorbent and adsorbate. The amount of dye removal at equilibrium increased from 3.02 mg/g to 3.036 mg/g for Rd6G with the slightly increase in dye concentration at different temperature 32^0, 40^0 and 50 °C (Fig. 8).

Adsorption kinetic studies

The adsorption kinetics data which were obtained from the experiments were analyzed in terms of pseudo- first order, pseudo - second order and Elovich models. Results obtained from the calculations (Table 1) reveal that the correlation coefficient R^2 values for the pseudo - second order model (0.999, 0.999, 0.999 and 0.999) were larger in comparison to the R^2 from the first - order model (0.835, 0.864, 0.940 and 0.942) and the estimated q_e values are observed to be in good agreement with experimental values of the adsorbent q_e (Fig. 9). The usability of this model suggest that the rate limiting step may be chemisorption involving valancy forces via sharing or exchange of electrons between adsorbent and adsorbate (Ho and McKay, 1998). This is in accord once with the studies of Rd6G adsorption onto candle shoot (Virendra Singh and Vaish, 2018), *Clitoria fairchildiana* (CF) pods as low cost adsorbent (Silva et al., 2020), Graphite (Nagaraju sykam et al., 2019) and Magnetic Graphene oxide (Gautam et al., 2020). Rh6G of pseudo - second order kinetic model equation can be represented by the following equation.

$$\frac{t}{q} = \frac{1}{K2{q_e}^2} + \frac{t}{q_e}, \tag{1}$$

where "$K2$" is the rate constant of pseudo second order sorption (mg/g), 'qe' is the amount of Rh6G adsorbed at equilibrium and 't' is time required.Adsorption isotherms

The adsorption isotherm demonstrates the particular relation between the concentration of adsorbate and its degree of collection onto adsorbent surface at constant temperature. Entire Experiments were carried out at room temperature (32 °C). The basic assumption of adsorption isotherm is that every adsorption site is equivalent and the binding ability of a particle is independent of whether or not adjacent sites are occupied (Shukla and Pai, 2005; Navneet and Kaur, 2017). Three isotherm models were used to fit the adsorption data i.e., Langmuir, Freundlich and Dubinin-Radushkevich (DR). These parameters are tabulated (Table 2). Langmuir isotherm expressed monolayer adsorption on to surface. This model suppose adsorbate molecules can only gathers on one localized site without lateral interaction between the adsorbed molecules, even on the nearby sites. As well, it gathers no further adsorption that can occur after equilibrium. Langmuir isotherm recommended the adsorption is favorable or not using the dimensionless constant separation factor. R_L value indicates adsorption isotherm is favorable, unfavorable or linear adsorption. Rh6G shows very well fitted to the Langmuir isotherm model (Fig. 10) and its equation can be explained by the following equation.

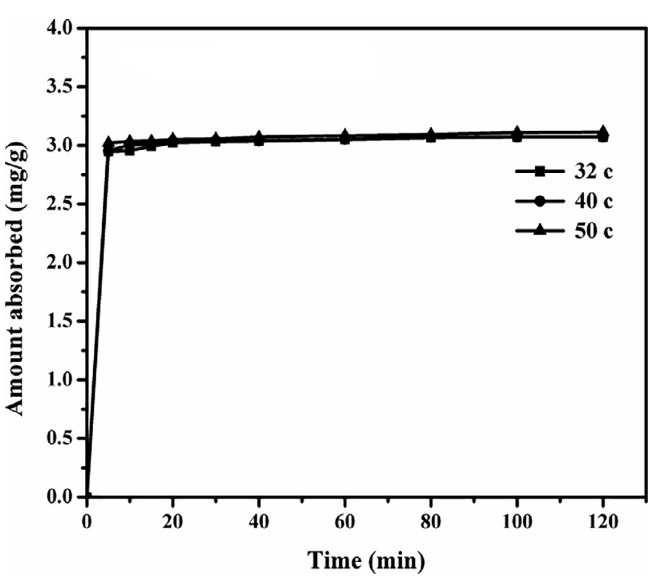


Fig. 8. Effect of different temperature on Rhodamine 6G.

Table 1
Kinetic parameters for the adsorption of Rh6G onto the *Litchi chinensis* peel.

Model	Parameter	Rhodamine 6G
Langmuir	Q_o (mg/g)	6.666
	B (L/mg)	0.465
	R^2	0.9998
Freundlich	K_f (mg $^{1-1/n}$ L $^{1/n}$ g $^{-1}$)	2.66
	N	3.149
	R^2	0.9543
Dubinin-Radushkevich	q_m (mg/g)	26.703
	E (KJ/mol)	50
	R^2	0.9642

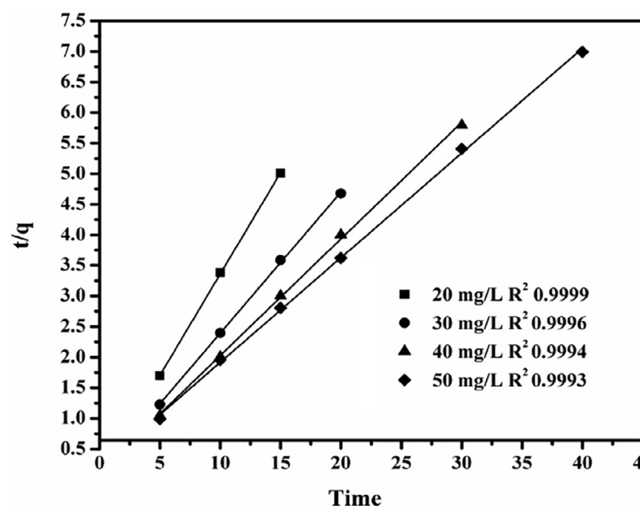


Fig. 9. Second-order kinetics plots for the removal of Rhodamine 6 G.

Table 2
Isotherm parameters for the adsorption of Rh6G onto the *Litchi chinensis* peel.

Adsorption	Q max (mg/g)	References
Coffee ground power	17.369	(Shen and Gondal, 2017)
Na+- Montmorillonite	0.4	(Gemeay 2002)
Biological sludge	16.3	(Annadurai et al., 2003)
Trichoderma harzianum mycelial waste	3.4	(Sadhasivam et al., 2007)
Graphene oxide		
Activated carbon	23.3	(Ren et al., 2014)
Clitoria fairchildiana	6.4	(Annadurai et al., 2001)
Almond shell (*Prunusdulcis*)	73.84	(Silva et al., 2020).
Graphene–sand	32.6	(Senturk et al., 2010)
Palm shell powder	55	(Gupta et al., 2012)
Cellulose acetate-triethoxysilane	105	(Sreelatha & Padmaja 2008)
Litchi Chinensis	48.62	(Leong et al., 2019)
	6.666	Present work

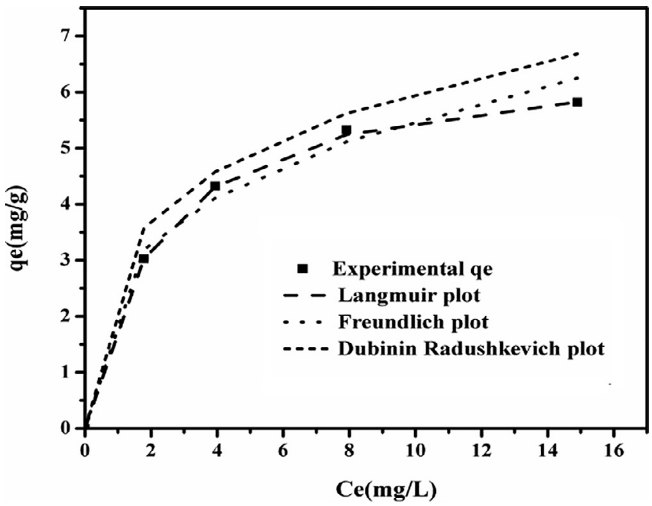


Fig. 10. Comparison of adsorption isotherms of Rhodamine 6 G.

$$\frac{C_e}{Q_e} = \left(\frac{1}{Q_0\,\mathrm{b}}\right) + \left(\frac{C_e}{Q_0}\right) \tag{2}$$

where C_e is the equilibrium concentration of Rd6G dye in the solution (mg/L), Q_e is the amount of dye adsorbed per unit weight of adsorbent (mg/g litchi peel), Q_0 is the amount of adsorption capacity (mg/g), b is a constant that relates to the heat of adsorption (L/mg). Freundlich isotherm used to illuminates the adsorption characteristics for the heterogeneous system. Dubinin-Radushkevich isotherm explains the adsorption mechanism with energy distribution onto a surface. This is in accord to studies of Rd6G adsorption onto acid treated banana peel (Oyekanmi et al., 2019), candle shoot (Singh and Vaish, 2018), $ZnFe_2O_4$ nano composite (Konicki et al., 2017). Some adsorbents of adsorption capacities of Rd6G are reported (Table 3).

Effectct of pH adsorption studies

Adsorption and desorption is the most important parameter. Effect of initial solution pH was investigated using various pH values in the range of 2 – 9. The initial Rd6G dye concentration of 20 and 30 mg/L and the removal rate decreased from 90.9 to 86.3 % for 30 mg/L of Rd6G concentration for *Litchi chinensis*. The adsorption of cationic dye percent removal of pH 8 shows the higher percent removal compared to other pH (Fig. 11). The solutions of pH were adjusted to the required pH by adding drops of 0.1 M NaOH or 0.1 M HCl. The zero point charge (pH_{zpc}) of *Litchi chinensis* was found to be 7.0. The adsorption sites on the adsorbent less than pH 7.0 is positive in nature and greater than 7.0 is negative in nature. With decrease of the solution pH, positive surface charge molecules of *Litchi chinensis* get attracted with negative charge of Rd6G molecules under electrostatic force of attraction and it favors greater adsorption of Rd6G. Similar findings are seen in *Clitoria fairchildiana* (CF) pods as low cost adsorbent (Silva et al., 2020), *Litchi chinensis* peel is a novel treatment (Hinge et al., 2016; Sirisha et al., 2019).

Table 3
Adsorption capacity comparison of the *Litchi chinensis* peel in this study with other adsorbents to remove Rh6G.

Adsorption	Q max (mg/g)	References
Coffee ground power	17.369	(Shen and Gondal, 2017)
Na+- Montmorillonite	0.4	(Gemeay 2002)
Biological sludge	16.3	(Annadurai et al., 2003)
Trichoderma harzianum mycelial waste	3.4	(Sadhasivam et al., 2007)
Graphene oxide		
Activated carbon	23.3	(Ren et al., 2014)
Clitoria fairchildiana	6.4	(Annadurai et al., 2001)
Almond shell (*Prunusdulcis*)	73.84	(Silva et al., 2020)
Graphene–sand	32.6	(Senturk et al., 2010)
Palm shell powder	55	(Gupta et al., 2012)
Cellulose acetate-triethoxysilane	105	(Sreelatha & Padmaja 2008)
Litchi Chinensis	48.62	(Leong et al., 2019)
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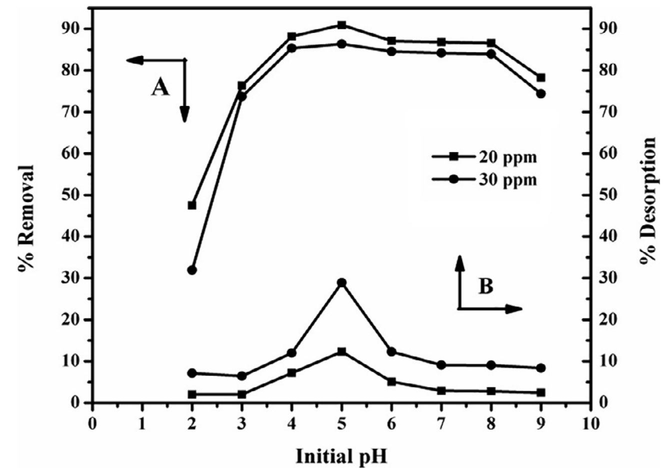

Fig. 11. Effect of pH on adsorption and desorption of Rhodamine 6 G.

Desorption studies

Desorption studies show that recycling of the spent adsorbent and the recovery of the Rd6G. would make the treatment process more economical. In complete desorption of Rd6G at different concentration suggests that the chemisorptions was the major mode of adsorption. Similar results are observed in *Sapindus emarginatus* (Jasper and Sumithra, 2020) *Litchi chinensis* (Sirisha et al., 2019), magnetic iron oxide (Ranjbari et al., 2015).

Studies of thermodynamics of adsorption

Thermodynamic parameters for the adsorption of Rd6G on *Litchi chinensis* calculated by the following equation (Table 4).

$$\Delta G^{\circ} = -RT\ \ln K_c \tag{3}$$

where R is gas constant (8.314 J/mol/K), and T is the temperature in (K):

$$\log K_c = (\Delta S^{\circ}/2.303 \times R) - (\Delta H^{\circ}/2.303 \times RT) \tag{4}$$

The values of ΔH^o and ΔS^o were determined from the slope and intercept of van't Hoff plots log $K_c$vs 1/T. Negative values of ΔH^o suggest the exothermic nature of adsorption and ΔG^o (< -40 kJ mol^{-1}) indicate spontaneous and favorable nature of adsorption and beneficial activity under the experimental

Table 4
Thermodynamic parameters for the adsorption of Rh6G on to the *Litchi chinensis*.

Temperature	K_c	ΔG^o (KJ/mol)	ΔH^o (KJ/mol)	ΔS^o (KJ/mol)
30	1.699	−1.344	−2.048	11.141
40	1.743	−1.446		
50	1.778	−1.545		

conditions. It involved in physical adsorption. The positive value of ΔS° with *Litchi chinensis* indicates increased randomness at the solid solution interface. Similar findings are seen in Water treatment sludge (Rashed et al., 2016) *Clitoria fairchildiana* (CF) pods as low cost adsorbent (Silva et al., 2020) Olive tree pruning waste (Anastopoulos et al., 2018).

Conclusion

The study noticed that *Litchi chinensis* acts as a good adsorbent for the removal of Rd6G from aqueous solutions. The adsorption kinetics data followed pseudo-second order kinetic model with high correlation coefficient almost reaching a unit value for Rd6G. The removal as a result increases with increase in adsorbent dosage. The optimum operational pH was evaluated as 5.0 of Rd6G. The use of waste *Litchi chinensis* peel can be successfully taken in treating large amount of dye wastewaters.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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